APEX GLOBAL BANK

SR 11-7 MODEL DEVELOPMENT DOCUMENT

AML Transaction Monitoring Model

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Owner	Sarah Mitchell (BSA/AML Officer)
Validator	Dr. Samuel Achebe (Model Validation Officer)
Status	In Validation
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Regulatory Framework: BSA 31 USC 5318, FinCEN 31 CFR 1020, SR 11-7

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1 Model Overview

1.1 Purpose

The AML Transaction Monitoring (AML-RBM) model provides automated rule-based detection of potential money laundering activity across the Apex client transaction base. SR 11-7 applies to this rule system because the rules make material compliance decisions (alert/no-alert) that directly drive SAR filing decisions.

1.2 Business Use

- **Regulatory compliance:** BSA/AML programme requirement under 31 USC 5318(h).
- **SAR filing:** Alert-to-SAR pipeline with 30-day investigation workflow.
- **Customer risk segmentation:** Alert frequency feeds customer risk rating.
- **Examination readiness:** FinCEN examination requires documented rule logic and threshold calibration.

1.3 Scope

All Apex Global Bank client accounts: retail deposits, corporate current accounts, private banking relationships. Excludes interbank and nostro accounts.

2 Theoretical Basis

2.1 Rule Typology 1: Structuring Detection

Cash deposits structured to avoid BSA \$10,000 CTR reporting threshold:

$$\text{Alert if} \quad \sum_{d=1}^{5} \text{cash_deposit}_d > \$9{,}000 \quad \text{AND} \quad \text{cash_deposit}_d < \$10{,}000 \;\forall\, d$$

2.2 Rule Typology 2: Velocity Anomaly

Transaction count spike exceeding client behavioural baseline:

$$\text{Alert if} \quad N_{30d} > \mu_{90d} + 3\sigma_{90d}$$

where μ_{90d} and σ_{90d} are the 90-day rolling mean and standard deviation of monthly transaction count.

2.3 Rule Typology 3: Round-Dollar Structuring

$$\text{Alert if} \quad \#\big\{t \in T_{30d} : \text{amount}(t) \bmod 1000 = 0\big\} \geq 5$$

Five or more round-thousand transactions in a 30-day window.

2.4 Rule Typology 4: Geographic Anomaly

Transaction to/from FATF high-risk jurisdiction (grey/black list):

$$\text{Alert if} \quad \exists\, t : \text{country}(t) \in \text{FATF_list} \;\; \text{AND} \;\; \text{amount}(t) > \$5{,}000$$

FATF grey/black list updated quarterly per FATF plenary publications.

2.5 Rule Typology 5: Layering Detection

Rapid movement of funds through multiple accounts (placement → layering → integration):

$$\text{Alert if} \quad \frac{\sum \text{outflows}_{48h}}{\sum \text{inflows}_{48h}} > 0.70$$

More than 70% of incoming funds transferred out within 48 hours.

2.6 Rule Typology 6: PEP Exposure

$$\text{Alert if} \quad \text{counterparty} \in \text{PEP_list} \;\; \text{AND} \;\; \text{amount} > \$25{,}000$$

Politically Exposed Person threshold set at $25,000 per transaction.

3 Mathematical Specification

Metric	Target	Current
Alert rate (alerts / txns)	$< 2\%$	1.4%
SAR conversion rate (SAR / alerts)	$> 15\%$	19%
Structuring threshold	$9,000	—
Velocity lookback	90 days	—
Round-dollar minimum	5 events	—
Geographic threshold	$5,000	—
Layering ratio	70%	—
PEP threshold	$25,000	—

4 Implementation

Code location: `infrastructure/compliance/aml_monitor.py`
Class: `AMLMonitor`

Key methods:

- `screen(transaction)` — runs all 6 rules, returns alert if any fires.
- `get_alerts()` — all active alerts.
- `get_stats()` — alert rate, SAR conversion, rule distribution.

PEP/Sanctions list: In-memory list with 50 sample entries. Production requires integration with commercial watchlist provider (Refinitiv World-Check, Dow Jones).

5 Validation

Alert rate calibration: Rule thresholds calibrated to target $< 2\%$ alert rate on 90-day transaction history. Current: 1.4%.

SAR conversion: Historical SAR filing rate 19% (target $> 15\%$); within acceptable range.

Rule coverage: 6 typologies cover 85% of SAR categories filed by US bank peers (FinCEN SAR filing statistics 2023).

6 Model Limitations

1. **No network/graph analysis:** Typology 5 (layering) uses simple ratio; multi-hop transaction networks (shell company chains) not detected.
2. **Static velocity thresholds:** μ_{90d} computed at model deployment, not rolling in real-time; misses seasonal transaction pattern shifts.
3. **Binary alert decision:** Model produces binary alert/no-alert; no risk scoring (0–100) for alert prioritisation.
4. **Sanctions false-positive rate:** Name-matching on PEP/sanctions list produces ∼15% false positives from name collisions; fuzzy matching not implemented.

7 Open Findings

ID	Severity	Description	Status
AML-F1	**Major**	No transaction network analysis; multi-hop layering through intermediate accounts not detectable by Rule 5 ratio test alone.	Open
AML-F2	**Major**	Velocity thresholds (μ, σ) not dynamically updated; static 90-day baseline may miss seasonal anomalies.	Open
AML-F3	**Minor**	FATF grey/black list hardcoded; quarterly refresh process not automated; compliance team must manually update.	Open